Binary Parsers in Typescript

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Binary Parsers

1. Varian NMR data parser
2. Development Workflow
3. Related Projects

Binary Parsers

What are binary parsers?

Binary Parsers

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Why not Python?

Binary Parsers

What are binary parsers?

Why not Python?

Companies using these tools

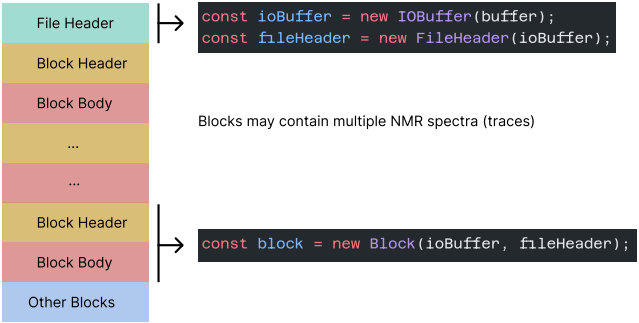
Zakodium/NMRium

https://nmrium.org/about

Programming Languages

- **Typescript**
  - For Node and Browser
  - Different compilation targets.
  - TF.js and ONNX
- Python
  - TF or PyTorch
  - Numpy and other libraries
- Bash
  - automation and scripts

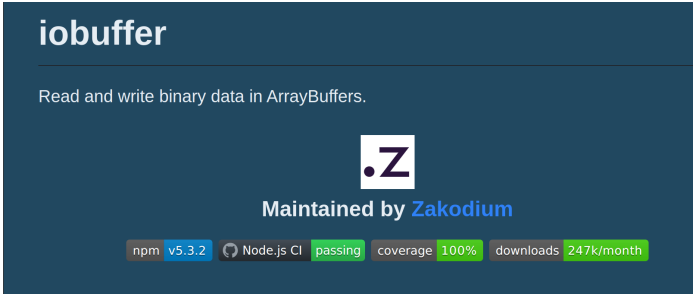
Varian Converter



Convert Varian NMR data to JSON

IOBuffer

- IOBuffer: Read and write binary data in memory



https://github.com/image-js/iobuffer

IOBuffer

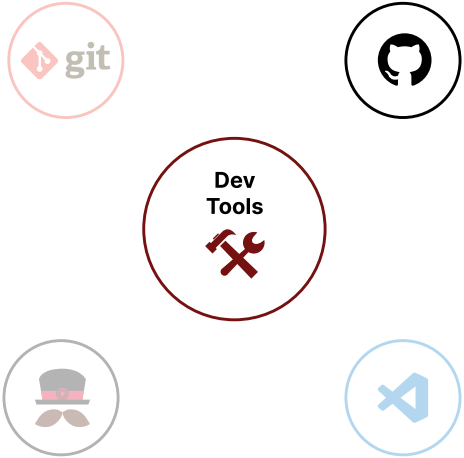
```
import { IOBuffer } from 'iobuffer';

const io = new IOBuffer()// offset 0
io.writeChars('Hello world') // offset now 11
.writeUint32(42)
.setBigEndian()
.writeUint32(24)
.skip(2)
.rewind() // back to 0
.readArray(11,"uint8") // read characters at once
```

Contributions: text decoder, readArrays of typed-data, host-endianness detection.

Development Workflow

Development Workflow

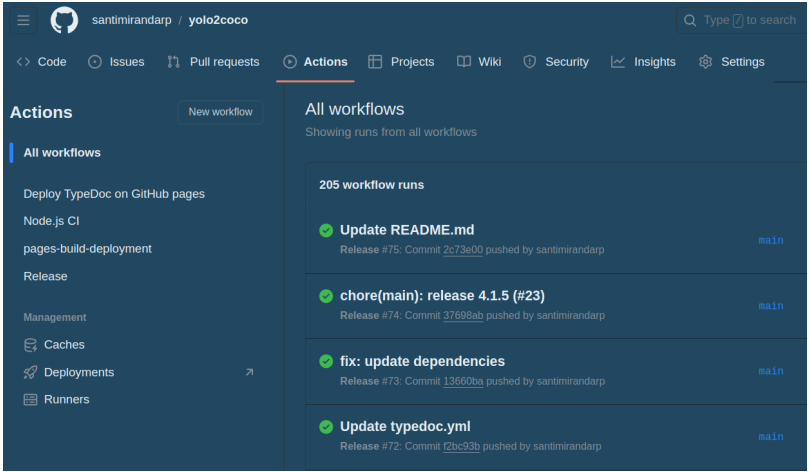


Github & VSCode

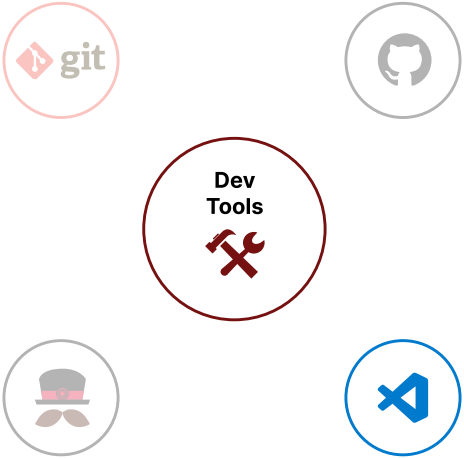
VS Code and Workflows (CI/CD)

Tool	Description
Semantic Release	changelog, publish
Jest, Vitest	testing
ESLint, Prettier	lint and format
Typescript	type-check
Typedoc	docs

Workflows


santimirandarp / yolo2coco
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
Actions
New workflow
All workflows
Deploy TypeDoc on GitHub pages
Node.js CI
pages-build-deployment
Release
Management
Caches
Deployments
Runners
All workflows
Showing runs from all workflows
205 workflow runs
Update README.md
Release #75: Commit 2c73e00 pushed by santimirandarp
main
chore(main): release 4.1.5 (#23)
Release #74: Commit 37698ab pushed by santimirandarp
main
fix: update dependencies
Release #73: Commit 13660ba pushed by santimirandarp
main
Update typedoc.yml
Release #72: Commit f2bc93b pushed by santimirandarp
main

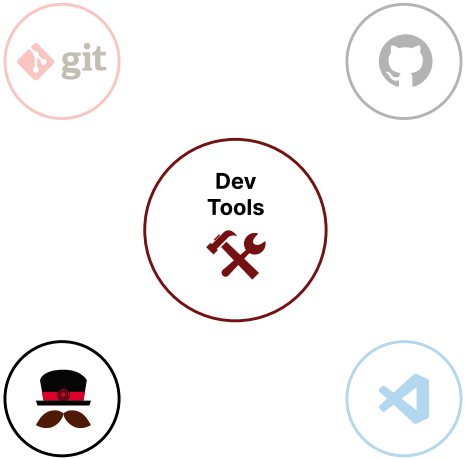
Development Workflow



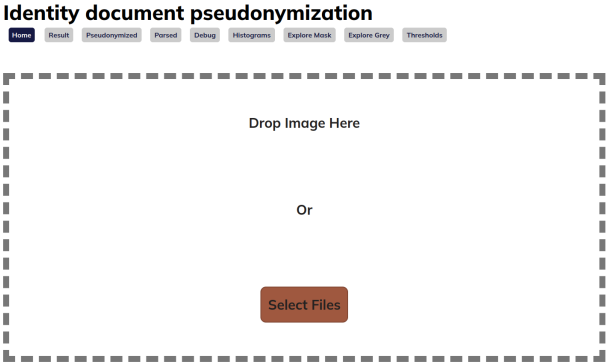
VSCode

- Remote SSH
- NodeJS Debugger
  - ☹ Configure `launch.json`
  - ☺ Track all variables and find bugs

Development Workflow

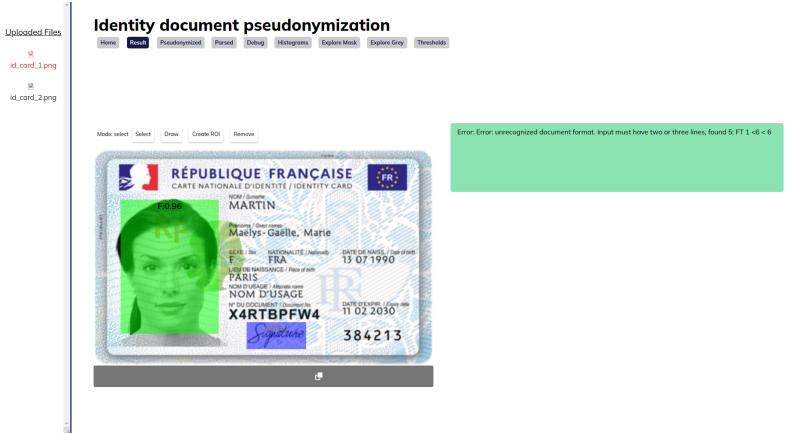


Other Projects



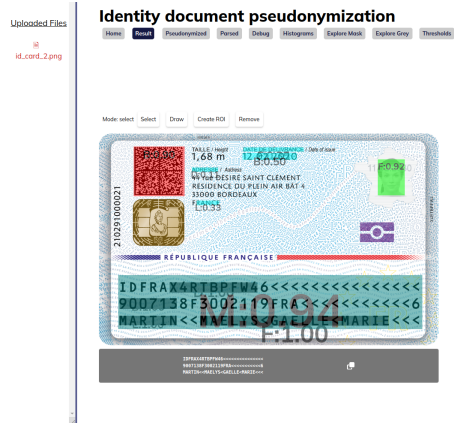
Pseudonymisation of Identity Documents

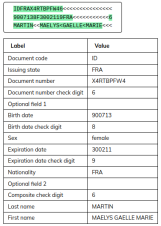
Other Projects



mrz.zakodium.com

Other Projects





IDFRAX4RTBPFW46<<<<<<<<<<<<<<<
9007138F3002119FRA<<<<<<<<<<<6
MARTIN<<MAELYS<GAELLE<MARIE<<<

Label	Value
Document code	ID
Issuing state	FRA
Document number	X4RTBPFW4
Document number check digit	6
Optional field 1	
Birth date	900713
Birth date check digit	8
Sex	female
Expiration date	300211
Expiration date check digit	9
Nationality	FRA
Optional field 2	
Composite check digit	6
Last name	MARTIN
First name	MAELYS GAELLE MARIE

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Other Projects

- **Pseudonymisation of Identity Documents**

  Deep Learning project (private repository)
- **nnls** (Non-negative least squares)

  https://github.com/mljs/nnls
- **fcnnls** (Fast Combinatorial Non-negative least squares)

  https://github.com/mljs/fcnnls
- **polynomial-regression**

  https://github.com/mljs/polynomial-regression
- **spc-parser**

  https://github.com/cheminfo/spc-parser
- **wdf-parser**

  https://github.com/cheminfo/wdf-parser

Thanks!